F-REGAL: Federated Graph Alignment

Proof of Concept with PubMed, REGAL, and FedAvg Integration

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Problem and Goal

- In federated graph learning, clients cannot share raw graphs, so cross-silo node correspondence is hard.
- Existing FL methods can train models, but do not directly solve node-to-node alignment.
- We target a practical question: can we align nodes across private silos while preserving useful privacy constraints?

Goal: Build a baseline pipeline that performs federated alignment and supports downstream GNN training.

Implementation Difficulties

- **No raw graph sharing:** privacy policies block sharing edges and local neighborhoods.
- **Heterogeneous silos:** clients have different graph slices and label distributions.
- **Alignment requirement:** downstream learning improves if equivalent nodes are matched across silos.
- **Tradeoff:** stronger privacy protection can distort alignment signal.

Method Summary: F-REGAL

- **Local structural identity:** each client computes k-hop degree-based fingerprints.
- **Shared latent space:** clients map fingerprints to a common basis using xNetMF-style similarity projection.
- **Server alignment:** nearest-neighbor search (KD-tree) produces cross-client node matches.
- **Privacy mechanism:** optional Laplacian noise added to shared embeddings before alignment.

Core projection form used in notebook:

$$Y = CU\sqrt{\Sigma}$$

Client and Server Roles

Client-side (private)

- Build local adjacency-derived descriptors.
- Combine structural and attribute features.
- Project to shared space using server basis.
- Optionally add Laplacian DP noise.

Server-side (coordinator)

- Set global config and landmark basis.
- Broadcast projection metadata.
- Receive only transformed representations.
- Run KD-tree alignment and report matches.

Implementation: Dataset, Client, Server

Dataset

- PubMed from `Planetoid`
- Undirected NetworkX graph
- Two client subgraphs sampled at 85% node coverage each

Client

- Extract structural identity ($K = 2$, $\delta = 0.5$)
- Fuse structure + attributes
- Build local similarity matrix C
- Optional DP noise ($b = 0.05$)

Server

- Build basis from 250 landmarks
- Compute global projection parameters (U, Σ)
- Align embeddings with top-1 nearest neighbor

Configuration Used

- Structural extraction depth and decay: $K = 2$, $\delta = 0.5$.
- Shared landmark basis size: 250 proxy nodes.
- Two privacy settings tested:
  - Utility bound: DP scale $= 0.0$.
  - Privacy bound: DP scale $= 0.05$.
- Alignment metric: top-1 nearest-neighbor correctness on overlapping nodes.

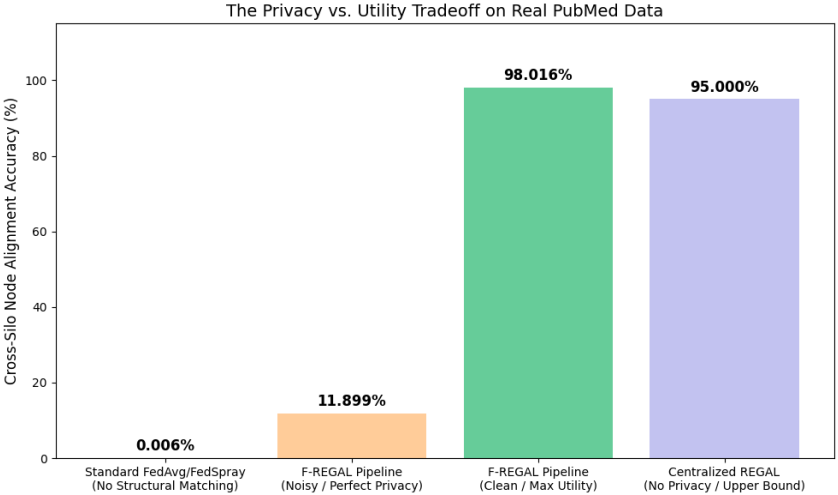
Experiment Design

- **Scenario A (Utility Bound):** no noise, clients share clean projected representations.
- **Scenario B (Privacy Bound):** Laplacian noise injected before sharing.
- Evaluate alignment only on true overlap nodes across client subgraphs.
- Then feed aligned representations into a two-layer GCN and run FedAvg rounds.

FedAvg settings used in notebook:

- 30 rounds, 5 local epochs, Adam with weight decay.

Experiment Chart



Results: Alignment Quality

Condition	Observed Outcome
Standard FL without explicit matching	Near-random alignment baseline
F-REGAL (no noise)	**98% alignment accuracy**
F-REGAL (+ Laplacian noise)	Noticeable drop from clean setting
Centralized REGAL reference	High alignment (used as upper-bound context)

Result: F-REGAL solves the federation gap in clean settings, but privacy protection introduces a utility tax.

Results: Downstream FedAvg Signal

- Aligned representations were used as node features in a two-layer GCN.
- Federated averaging ran for 30 rounds with local client updates.
- Global model performance was tracked on client-local blind test partitions.
- This shows alignment is not only measurable, but useful for downstream learning workflows.

Interpretation: Why Privacy Is Not Free

- Clean embeddings preserve fine structural signal, enabling near-centralized matching quality.
- DP noise improves resistance to reconstruction/triangulation attacks.
- The same noise perturbs nearest-neighbor geometry, reducing alignment precision.

Practical objective is a tuned midpoint: enough blur for privacy, enough signal for alignment and learning.

Shortcomings and Baseline Status

Current limitations

- This is a baseline proof of concept, not a fully production-hardened privacy system.
- Shared embeddings can still leak information under strong adversarial assumptions.
- Privacy is improved, but full formal end-to-end privacy guarantees are not yet ensured.
- Results are from one main dataset setup and should be stress-tested under broader non-IID settings.

Threat Model and Privacy Gap

- Assumed defense: embedding perturbation via Laplacian noise.
- Remaining risk: representation-level attacks may still infer sensitive structure.
- Missing today: formal privacy accounting tied to a concrete attacker capability model.

Key statement for this project: this work is a proof of concept baseline and does **not** guarantee full privacy.

Conclusion and Next Steps

- F-REGAL demonstrates feasible federated node alignment with meaningful empirical gains.
- The project confirms a core tradeoff: stronger privacy noise lowers matching accuracy.
- Immediate next steps:
  - calibrate DP across multiple scales and report full privacy-utility curves,
  - extend evaluation across more datasets and client partitions,
  - add stronger privacy accounting and attack-model testing.

Questions?

Thank you.

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